Assignment #1

1. (a) $S_e(t) = 0.5[s(t) + s(-t)] = 0.5[e^{\alpha_0 t} + e^{-\alpha_0 t}]$, $S_o(t) = 0.5[s(t) - s(-t)] = 0.5[e^{\alpha_0 t} - e^{-\alpha_0 t}]$

$\therefore s(t) = S_e(t) + S_o(t) = e^{\alpha_0 t}$

(b) $S_e(t) = 0.5[s(t) + s(-t)]$, $S_o(t) = 0.5[s(t) - s(-t)]$

$$S_e(t) = \begin{cases} 0.5 \cdot \sin(2t+3), & t>0 \\ 0.5 \cdot \sin(-2t+3), & t<0 \end{cases} \qquad S_o(t) = \begin{cases} 0.5 \cdot \sin(2t+3), & t>0 \\ -0.5\sin(-2t+3), & t<0 \end{cases}$$

$\therefore s(t) = S_e(t) + S_o(t)$

$$= \begin{cases} \sin(2t+3), & t>0 \\ 0, & \text{otherwise.} \end{cases}$$

2. a) $x(t) = \sin(\frac{\pi}{3}t) + 2\cos(\frac{8\pi}{3}t)$

Let $x_1(t) = \sin(\frac{\pi}{3}t) \Rightarrow T_1 = \frac{2\pi}{\frac{\pi}{3}} = 6$

$x_2(t) = 2\cos(\frac{8}{3}\pi t) \Rightarrow T_2 = \frac{2\pi}{\frac{8}{3}\pi} = \frac{3}{4}$

$\frac{T_1}{T_2} = \frac{6}{\frac{3}{4}} = 8$. $\therefore$ It's periodic with $T = 8T_2 = 6$.

b). $x(t) = \exp(j\frac{7}{6}\pi t) + \exp(\frac{5}{6}\pi t)$

Let $x_1(t) = \exp(j\frac{7}{6}\pi t) \Rightarrow T_1 = \frac{2\pi}{\frac{7}{6}\pi} = \frac{12}{7}$

$x_2(t) = \exp(\frac{5}{6}\pi t) \Rightarrow$ It's an aperiodic signal.

So $x(t)$ is an aperiodic signal.

c). $x(t) = 2\sin(\frac{3}{8}\pi t) + \cos(\frac{3}{4}t)$

Let $x_1(t) = 2\sin(\frac{3}{8}\pi t) \Rightarrow T_1 = \frac{2\pi}{\frac{3}{8}\pi} = \frac{16}{3}$

$x_2(t) = \cos(\frac{3}{4}t) \Rightarrow T_2 = \frac{2\pi}{\frac{3}{4}} = \frac{8}{3}\pi$

$\frac{T_1}{T_2} = \frac{\frac{16}{3}}{\frac{8}{3}\pi} = \frac{2}{\pi}$: It's an aperiodic signal because the ratio is not a rational number.

3. a). $P = \frac{1}{T}\int_0^T |x(t)|^2 dt \Rightarrow$ periodic signal.

$= \frac{1}{T}\int_0^T A^2 |\sin^2(t)| dt$

$= A^2 \cdot \frac{1}{T}\int_0^T \sin^2(t)\, dt$

$= \frac{A^2}{2}\cdot\frac{1}{T}\int_0^T \frac{(1-\cos 2t)}{2} dt$

$= \frac{A^2}{2}\cdot\frac{1}{T}[T - \int_0^T \cos 2t\, dt]$

since $\int_0^T \cos 2t\, dt = \frac{1}{2}\sin 2t\Big|_0^T = 0$

$\therefore P = \frac{A^2}{2T}\cdot T = \frac{A^2}{2}$. It's a power signal.

b). $E = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_0^{+\infty} |\exp(-at)|^2 dt$

$= \int_0^{+\infty} \exp(-2at)\, dt$

$= \frac{1}{-2a}\cdot \exp(-2at)\Big|_0^{+\infty}$, $a > 0$

$= -\frac{1}{2a}\cdot[0 - 1] = \frac{1}{2a}$

It's an energy signal.

c). $E = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |A\exp(bt)|^2 dt$, $b>0$

$= A^2 \int_{-\infty}^{+\infty} \exp(2bt)\, dt$, $b>0$

$= \frac{A^2}{2b} \cdot \exp(2bt)\Big|_{-\infty}^{+\infty}$

$= \frac{A^2}{2b} \cdot [+\infty - 0] = \infty$



$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$

$= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} A^2 \exp(2bt)\, dt$, $b>0$

$= \frac{A^2}{2} \cdot \lim_{T\to\infty} \frac{\left[\int_{-T}^{T} \exp(2bt)dt\right]'}{(T)'}$, $b>0$

$= \frac{A^2}{2} \cdot \lim_{T\to\infty} [\exp(2bT) - \exp(-2bT)]$, $b>0$

$= \frac{A^2}{2} \cdot [+\infty - 0] = +\infty$.



Note: $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{f'(x)}{g'(x)}$ is used in the above calculation.

d). So it's neither an energy nor a power signal

$E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$

$= \int_{0}^{+\infty} |\exp(-(a+jb)t)|^2 dt$, $a>0$

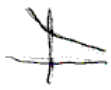
$= \int_{0}^{+\infty} |\exp(-at)\cdot\exp(-jbt)|^2 dt$

$= \int_{0}^{+\infty} |\exp(-at)|^2 \cdot |\exp(-jbt)|^2 dt$

$= \int_{0}^{+\infty} \exp(-2at) \cdot |\cos(-bt)+j\sin(-bt)|^2 dt$

$= \int_{0}^{+\infty} \exp(-2at) \cdot \left(\sqrt{\cos^2(-bt)+\sin^2(-bt)}\right)^2 dt$

$= \int_{0}^{+\infty} \exp(-2at)dt = \frac{1}{-2a} \cdot [\exp(-2at)\Big|_{0}^{+\infty}] = \frac{1}{2a}$



So it's an energy signal.